

Optimization for Machine Learning

CS-439

Lecture 10: Lower Bounds and Accelerated Gradient Descent

Martin Jaggi

EPFL – github.com/epfml/OptML_course

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Can We be Faster?

The rates of convergence of Gradient Descent for various function classes we have seen so far

Function Class	Rate for GD
Convex & L -Lipschitz	$\mathcal{O}(1/\sqrt{T})$
Convex & L -Smooth	$\mathcal{O}(1/T)$
μ -Strongly convex & L -Smooth	$\mathcal{O}(\exp\{-\mu T/L\})$

Table: Rates of convergence of Gradient Descent after T iterations for various function classes.

Question: Is GD the optimal algorithm ? What optimal rates of convergence do we expect for these class of functions?

Lower Bounds

Why do we need computational lower bounds?

- ▶ Lower bounds establish the optimality of an algorithm
- ▶ Lower bounds can guide the design of efficient algorithms

The lower bounds we present first appeared in the works of Nemirovski and Yudin [NY83] and follows the revised presentation by Nesterov [Nes18].

The Class of Algorithms

Before we start, let us define the class of gradient-based algorithms we are interested in.

Definition: For a function f , a **gradient-based algorithm** initialized at $\mathbf{x}_0$ is defined as a sequence of points $\mathbf{x}_0, \mathbf{x}_1, \dots$ generated by the following update rule:

$$\mathbf{x}_{t+1} \in \mathbf{x}_0 + \text{Span} \{ \nabla f(\mathbf{x}_0), \nabla f(\mathbf{x}_1), \dots, \nabla f(\mathbf{x}_t) \}$$

Note that **GD** with a constant stepsize η belongs to the class of gradient-based algorithm. By unraveling the updates of GD,

$$\mathbf{x}_t = \mathbf{x}_0 - \eta \sum_{i=0}^{t-1} \nabla f(\mathbf{x}_i),$$

belongs to the span of the past gradients.

The Strategy for Lower Bounds

- ▶ Construct a **pathological family** of functions, such that any gradient-based method cannot do better than a certain rate.
- ▶ For this lecture, we will focus on the class of **convex** and **L-smooth** functions over $\mathbb{R}^d$.

The Worst Function in the World

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be the function defined as

$$f(\mathbf{x}) = \frac{L}{4} \left[\frac{1}{2} \mathbf{x}[1]^2 + \frac{1}{2} \sum_{i=1}^{d-1} (\mathbf{x}[i] - \mathbf{x}[i+1])^2 + \frac{1}{2} \mathbf{x}[d]^2 - \mathbf{x}[1] \right]. \quad (1)$$

With the tri-diagonal matrix $A \in \mathbb{R}^{d \times d}$ defined by

$$A = \begin{bmatrix} 2 & -1 & 0 & \cdots & \cdots & 0 \\ -1 & 2 & -1 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \cdots & 0 \\ 0 & 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 2 \end{bmatrix},$$

the function f can be alternatively written as

$$f(\mathbf{x}) = \frac{L}{8} \mathbf{x}^\top A \mathbf{x} - \frac{L}{4} \mathbf{x}[1].$$

Properties of the Worst Function

Convexity and Smoothness. Note that $\nabla^2 f(\mathbf{x}) = (L/4) \cdot A$. For any vector $\mathbf{h} \in \mathbb{R}^d$, note that

$$\mathbf{h}^\top A \mathbf{h} = \sum_{i=1}^{d-1} (h[i] - h[i+1])^2 + h[1]^2 + h[d]^2.$$

As $\mathbf{h}^\top A \mathbf{h} \geq 0$, we have that A is positive semi-definite. Thus, f is convex. To prove smoothness,

$$\begin{aligned} \mathbf{h}^\top A \mathbf{h} &= \sum_{i=1}^{d-1} (h[i] - h[i+1])^2 + h[1]^2 + h[d]^2 \\ &\leq 2 \sum_{i=1}^{d-1} [h[i]^2 + h[i+1]^2] + h[1]^2 + h[d]^2 \leq 4 \sum_{i=1}^d h[i]^2 = 4 \|\mathbf{h}\|^2. \end{aligned}$$

Therefore, $A \preceq 4I_d$ and f is L -smooth.

Optimum of the Worst Function

Gradient. The gradient of f is given by

$$\nabla f(\mathbf{x}) = \frac{L}{4} (A\mathbf{x} - \mathbf{e}_1),$$

where $\mathbf{e}_1 = (1, 0, \dots)$ is the first elementary basis vector of $\mathbb{R}^d$. The optimum $\mathbf{x}^*$ is obtained by solving the linear system

$$A\mathbf{x}^* = \mathbf{e}_1.$$

The solution is given by

$$\mathbf{x}^*[i] = 1 - \frac{i}{d+1} \quad \forall 1 \leq i \leq d.$$

The Trajectory of The Gradient-Based Algorithm

Lemma

Without loss of generality, we can assume that $\mathbf{x}_0 = 0$. Then, for $t \leq d$ the trajectory of the gradient-based algorithm for the function f defined in Eq. (1) satisfies

$$\mathbf{x}_t \in \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_t\},$$

where $\mathbf{e}_i$ is the i -th elementary basis vector of $\mathbb{R}^d$.

Proof. The proof follows from induction. Let, $\mathbf{x}_s \in \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_s\}$ for all $0 \leq s < t$. Note that

- $\nabla f(\mathbf{x}_s) = \frac{L}{4} (A\mathbf{x}_s - \mathbf{e}_1) \in \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{s+1}\}$, i.e, the gradient update is restricted to the first $s + 1$ coordinates.

Therefore, $\mathbf{x}_{s+1} \in \text{Span}\{\nabla f(\mathbf{x}_0), \nabla f(\mathbf{x}_1), \dots, \nabla f(\mathbf{x}_s)\}$ is also restricted to the first $s + 1$ coordinates. By induction, we have that $\mathbf{x}_t \in \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_t\}$.

Lower Bound

Lemma

For any gradient-based algorithm and for $t = (d-1)/2$, there exists a function f defined in Eq. (1) for which the following lower bound holds:

$$\min_{s \leq t} f(\mathbf{x}_s) - f(\mathbf{x}^*) \geq \frac{3L\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{32(t+1)^2}$$

Comment. The lower bound is valid only when the iterations are comparable with the dimension of the problem, i.e, $t < d/2$.

Proof. From the previous lemma, we have that $\mathbf{x}_t \in \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_t\}$. Using this fact,

$$f(\mathbf{x}_t) \geq \min_{\mathbf{x} \in \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_t\}} f(\mathbf{x}).$$

Denote $\mathbf{x}_t^* = \underset{\mathbf{x} \in \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_t\}}{\text{argmin}} f(\mathbf{x})$, be the minimizer of f over the span of the first t coordinates.

Lower Bound Continued ...

Note that the function f over the set $\text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_t\}$ is

$$f_t(\mathbf{x}) = \frac{L}{4} \left[\frac{1}{2} \mathbf{x}[1]^2 + \frac{1}{2} \sum_{i=1}^{t-1} (\mathbf{x}[i] - \mathbf{x}[i+1])^2 + \frac{1}{2} \mathbf{x}[t]^2 - \mathbf{x}[1] \right].$$

Using the same computation as the minimizer of f , we have that the minimizer of f_t is given by

$$\mathbf{x}_t^*[i] = \begin{cases} 1 - \frac{i}{t+1} & \text{for } 1 \leq i \leq t, \\ 0 & \text{for } t+1 \leq i \leq d \end{cases}$$

Combining the previous results,

$$f(\mathbf{x}_t) - f(\mathbf{x}^*) \geq f(\mathbf{x}_t^*) - f(\mathbf{x}^*). \quad (2)$$

Lower Bound Continued ...

Computing $f(\mathbf{x}^*)$ and $f(\mathbf{x}_t^*)$,

$$\begin{aligned} f(\mathbf{x}^*) &= \frac{L}{4} \left[\frac{1}{2} \mathbf{x}^*[1]^2 + \frac{1}{2} \sum_{i=1}^{d-1} (\mathbf{x}^*[i] - \mathbf{x}^*[i+1])^2 + \frac{1}{2} \mathbf{x}^*[t]^2 - \mathbf{x}^*[1] \right] \\ &= \frac{L}{4} \left[\frac{1}{2} \left(\frac{d}{d+1} \right)^2 + \frac{d-1}{2} \left(\frac{1}{d+1} \right)^2 + \frac{1}{2} \left(\frac{1}{d+1} \right)^2 - \frac{d}{d+1} \right] = -\frac{L}{8} \left[1 - \frac{1}{d+1} \right] \end{aligned}$$

Similarly, $f(\mathbf{x}_t^*) = -\frac{L}{8} \left[1 - \frac{1}{t+1} \right]$. Using Eq. (2), we have that,

$$\begin{aligned} f(\mathbf{x}_t) - f(\mathbf{x}^*) &\geq \frac{L}{8} \left[\frac{1}{t+1} - \frac{1}{d+1} \right], \\ \min_{s \leq t} f(\mathbf{x}_s) - f(\mathbf{x}^*) &\geq \frac{L}{8} \left[\min_{s \leq t} \frac{1}{s+1} - \frac{1}{d+1} \right] = \frac{L}{8} \left[\frac{1}{t+1} - \frac{1}{d+1} \right] \end{aligned}$$

Using the assumption that $d+1 = 2(t+1)$, $f(\mathbf{x}_t) - f(\mathbf{x}^*) \geq \frac{L}{16} \frac{1}{t+1}$. We gave a lower bound for the residual for any gradient-based algorithm.

Lower Bound Continued ...

Note that the rates are usually also depend on $\|\mathbf{x}_0 - \mathbf{x}^\star\|^2$, we compute that to give the final lower bound.

$$\begin{aligned}\|\mathbf{x}_0 - \mathbf{x}^\star\|^2 &= \|\mathbf{x}^\star\|^2 = \sum_{i=1}^d \left(1 - \frac{i}{d+1}\right)^2 \\&= \sum_{i=1}^d \left(1 - \frac{2i}{d+1} + \frac{i^2}{(d+1)^2}\right) = d - 2\frac{d(d+1)}{2(d+1)} + \frac{d(d+1)(2d+1)}{6(d+1)^2} \\&= \frac{d(2d+1)}{6(d+1)} \leq \frac{d+1}{3} = \frac{2(t+1)}{3}\end{aligned}$$

The final bound using the above computation writes,

$$\frac{f(\mathbf{x}_t) - f(\mathbf{x}^\star)}{\|\mathbf{x}_0 - \mathbf{x}^\star\|^2} \geq \frac{L}{16} \frac{1}{t+1} \cdot \frac{3}{2(t+1)} = \frac{3L}{32(t+1)^2}$$

Matching the Lower Bound: Acceleration for Smooth Convex Functions

Nesterov 1983 [Nes83, Nes18]: There is a gradient-based algorithm that achieves the rate of $\mathcal{O}(1/T^2)$ for T steps on smooth convex functions, and by the lower bound of Nemirovski and Yudin, this is a best possible algorithm!

The algorithm is known as (Nesterov's) accelerated gradient descent.

A number of (similar) optimal algorithms with other proofs for the rate of $\mathcal{O}(1/T^2)$ are known, but there is no well-established “simplest proof”.

Here: a recent proof based on [potential functions](#) [BG17]. Proof is simple but not very instructive (it works, but it's not clear why).

Nesterov's accelerated gradient descent

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex, differentiable, and smooth with parameter L . Choose $\mathbf{z}_0 = \mathbf{y}_0 = \mathbf{x}_0$ arbitrary. For $t \geq 0$, set

$$\begin{aligned}\mathbf{y}_{t+1} &:= \mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t) \\ \mathbf{z}_{t+1} &:= \mathbf{z}_t - \frac{t+1}{2L} \nabla f(\mathbf{x}_t) \\ \mathbf{x}_{t+1} &:= \frac{t+1}{t+3} \mathbf{y}_{t+1} + \frac{2}{t+3} \mathbf{z}_{t+1}.\end{aligned}$$

- ▶ Perform a “smooth step” from $\mathbf{x}_t$ to $\mathbf{y}_{t+1}$.
- ▶ Perform a more aggressive step from $\mathbf{z}_t$ to $\mathbf{z}_{t+1}$.
- ▶ Next iterate $\mathbf{x}_{t+1}$ is a weighted average of $\mathbf{y}_{t+1}$ and $\mathbf{z}_{t+1}$, where we compensate for the more aggressive step by giving $\mathbf{z}_{t+1}$ a relatively low weight.

Why should this work??

Nesterov's accelerated gradient descent: Error bound

Theorem

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex and differentiable with a global minimum $\mathbf{x}^*$; furthermore, suppose that f is smooth with parameter L . Accelerated gradient descent yields

$$f(\mathbf{y}_T) - f(\mathbf{x}^*) \leq \frac{2L \|\mathbf{z}_0 - \mathbf{x}^*\|^2}{T(T+1)}, \quad T > 0.$$

To reach error at most ε , accelerated gradient descent therefore only needs $\mathcal{O}(1/\sqrt{\varepsilon})$ steps instead of $\mathcal{O}(1/\varepsilon)$.

Recall the bound for gradient descent:

$$f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{L}{2T} \|\mathbf{x}_0 - \mathbf{x}^*\|^2, \quad T > 0.$$

Nesterov's accelerated gradient descent: The potential function

Idea: assign a **potential** $\Phi(t)$ to each time t and show that $\Phi(t+1) \leq \Phi(t)$.

Out of the blue: let's define the potential as

$$\Phi(t) := t(t+1) (f(\mathbf{y}_t) - f(\mathbf{x}^*)) + 2L \|\mathbf{z}_t - \mathbf{x}^*\|^2.$$

If we can show that the potential always decreases, we get

$$\underbrace{T(T+1) (f(\mathbf{y}_T) - f(\mathbf{x}^*)) + 2L \|\mathbf{z}_T - \mathbf{x}^*\|^2}_{\Phi(T)} \leq \underbrace{2L \|\mathbf{z}_0 - \mathbf{x}^*\|^2}_{\Phi(0)}.$$

Rewriting this, we get the claimed error bound.

(optional material) Potential function decrease: Three Ingredients

Sufficient decrease for the smooth step from $\mathbf{x}_t$ to $\mathbf{y}_{t+1}$:

$$f(\mathbf{y}_{t+1}) \leq f(\mathbf{x}_t) - \frac{1}{2L} \|\nabla f(\mathbf{x}_t)\|^2; \quad (3)$$

Vanilla analysis for the more aggressive step from $\mathbf{z}_t$ to $\mathbf{z}_{t+1}$: ($\gamma = \frac{t+1}{2L}$, $\mathbf{g}_t = \nabla f(\mathbf{x}_t)$):

$$\mathbf{g}_t^\top (\mathbf{z}_t - \mathbf{x}^\star) = \frac{t+1}{4L} \|\mathbf{g}_t\|^2 + \frac{L}{t+1} (\|\mathbf{z}_t - \mathbf{x}^\star\|^2 - \|\mathbf{z}_{t+1} - \mathbf{x}^\star\|^2); \quad (4)$$

Convexity (graph of f is above the tangent hyperplane at $\mathbf{x}_t$):

$$f(\mathbf{x}_t) - f(\mathbf{w}) \leq \mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{w}), \quad \mathbf{w} \in \mathbb{R}^d. \quad (5)$$

(optional material) Potential function decrease: Proof

By definition of potential,

$$\begin{aligned}\Phi(t+1) &= t(t+1)(f(\mathbf{y}_{t+1}) - f(\mathbf{x}^*)) + 2(t+1)(f(\mathbf{y}_{t+1}) - f(\mathbf{x}^*)) + 2L\|\mathbf{z}_{t+1} - \mathbf{x}^*\|^2, \\ \Phi(t) &= t(t+1)(f(\mathbf{y}_t) - f(\mathbf{x}^*)) + 2L\|\mathbf{z}_t - \mathbf{x}^*\|^2.\end{aligned}$$

Now, prove that $\Delta := (\Phi(t+1) - \Phi(t))/(t+1) \leq 0$:

$$\begin{aligned}\Delta &= t(f(\mathbf{y}_{t+1}) - f(\mathbf{y}_t)) + 2(f(\mathbf{y}_{t+1}) - f(\mathbf{x}^*)) + \frac{2L}{t+1} \left(\|\mathbf{z}_{t+1} - \mathbf{x}^*\|^2 - \|\mathbf{z}_t - \mathbf{x}^*\|^2 \right) \\ &\stackrel{(4)}{=} t(f(\mathbf{y}_{t+1}) - f(\mathbf{y}_t)) + 2(f(\mathbf{y}_{t+1}) - f(\mathbf{x}^*)) + \frac{t+1}{2L} \|\mathbf{g}_t\|^2 - 2\mathbf{g}_t^\top(\mathbf{z}_t - \mathbf{x}^*) \\ &\stackrel{(3)}{\leq} t(f(\mathbf{x}_t) - f(\mathbf{y}_t)) + 2(f(\mathbf{x}_t) - f(\mathbf{x}^*)) - \frac{1}{2L} \|\mathbf{g}_t\|^2 - 2\mathbf{g}_t^\top(\mathbf{z}_t - \mathbf{x}^*) \\ &\leq t(f(\mathbf{x}_t) - f(\mathbf{y}_t)) + 2(f(\mathbf{x}_t) - f(\mathbf{x}^*)) - 2\mathbf{g}_t^\top(\mathbf{z}_t - \mathbf{x}^*) \\ &\stackrel{(5)}{\leq} t\mathbf{g}_t^\top(\mathbf{x}_t - \mathbf{y}_t) + 2\mathbf{g}_t^\top(\mathbf{x}_t - \mathbf{x}^*) - 2\mathbf{g}_t^\top(\mathbf{z}_t - \mathbf{x}^*) \\ &= \mathbf{g}_t^\top((t+2)\mathbf{x}_t - t\mathbf{y}_t - 2\mathbf{z}_t) \stackrel{(\text{algo})}{=} \mathbf{g}_t^\top \mathbf{0} = 0. \quad \square\end{aligned}$$

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